Gaddam Mani Suchismith Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.66
Narayana Junior College	Class XII (TSBIE)	2023	96.60%
Narayana High School	Class X (SSC)	2021	10.00%

PROJECTS

Cost-Benefit & Sustainability Analysis: Automatic Water Faucets

- Analyzed water-usage data from a controlled two-phase faucet comparison study, quantifying 36% and 28% reductions in consumption in men's and women's restrooms, respectively, after switching from manual to automatic faucets.
- Built a multi-variable cost-benefit analysis incorporating water, sewer, installation, maintenance, and battery costs; recalculated the ROI using current market rates, reducing the estimated payback period from 74 to 15 years and demonstrating improved long-term economic viability and strengthening the case for adopting automatic faucets.
- Structured the faucet sustainability assessment around environmental, social, and economic dimensions, then mapped the findings to 2 UN Sustainable Development Goals: SDG 12 (Responsible Consumption & Production) and SDG 11 (Sustainable Cities & Communities), highlighting water conservation and more resilient urban infrastructure.

Delivery Operations & Rider Experience Analytics

- Cleaned and integrated 4 datasets (10,000 orders, restaurant wait times, driver contracts, weather and traffic) and engineered metrics like SLA gap, restaurant wait time and order hour to analyze the order-to-delivery journey.
- Found that batching 3+ orders caused 10-15 minute delays and that time-of-day congestion predicted delays better than weather or restaurant prep time. Recommended a dynamic tiered SLA (25/30/45 min), peak-hour batching limits and a hybrid pricing model, since distance alone did not explain order price across different delivery zones.
- Analyzed 11,849 rider reviews (2,513 after cleaning) and found that one-word reviews inflated star ratings, while detailed negative reviews pointed to earnings opacity, penalties and inconsistent incentives. Recommended live earnings dashboards and penalty-free cancellation quotas to improve rider trust.

Feasibility & Economic Analysis of Rural Water Supply Scheme

- Conducted a feasibility and demand assessment for a rural water-supply solution serving 6 habitations, projecting demand for 17,400 people by 2056 and identifying key infrastructure and service requirements.
- Evaluated cost-benefit and resource trade-offs for alternative pipeline capacities using whole-life economics, comparing capital investment against recurring energy costs to support infrastructure investment decisions.
- Identified operational sustainability opportunities, highlighting energy as the major recurring cost (~32 lakh/year) and assessing a 250 kWp solar PV option to reduce operating expenditure and improve long-term project viability.

SKILLS

Technical Skills: C/C++, Python, MySQL

Tools: Latex, MS Excel, QGIS, STAAD, Solid Edge

Languages: Telugu, Hindi, English

Soft Skills: Communication, Critical and Logical Thinking, Problem-solving, Adaptability

RELEVANT COURSES

Entrepreneurship: Introduction to Entrepreneurship, Innovation Management, Marketing Research, Competitive Strategy for Startup, Fundamentals of Organizational Structure, Visual Analytics, HR and Leadership, Text Analytics for Business

Mathematics: Calculus, Introduction to Probability, Transform Techniques, Complex Variables, Introduction to Statistics, Differential Equations, Elementary Linear Algebra.

POSITIONS OF RESPONSIBILITY

Civil Engineering Society (CES)- Design Executive:

Contributed to planning and execution of departmental events, creating engaging event designs and visual communication to enhance outreach and student participation, while developing creative problem-solving, event coordination, teamwork.

Cepheid (Astronomy Club)- Events and Outreach Coordinator:

Led ideation and execution of engaging astronomy initiatives, coordinating outreach and event operations for large-scale programs across schools and institutions, ensuring active participation and interactive learning.

Sunshine (The Councelling Cell of IITH)- Design member:

Created engaging and promotional designs for mental health and well-being initiatives, translating event themes into creative visual content to enhance campus outreach and awareness while developing creativity, visual communication.

EXTRACURRICULAR

- Active member of the **National Service Scheme (NSS), IIT Hyderabad**, contributing to community service and social outreach initiatives.
- Pursued photography as a personal interest, developing a basic understanding of camera settings, composition, and exposure through hands-on practice.
- Passionate about gardening and plant care, with a strong interest in nature, sustainable living, and nurturing plants.